Colin Aten

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Education

University of Delaware *2022-08 to 2026-05*
Bachelor of Science, Computer Engineering *GPA: 3.85 / 4.0*

- **Honors & Minors:** Honors College, Computer Science Minor, Cybersecurity Minor, German for Engineers Program, IEEE Honor Society Tau Beta Pi
- **Relevant Coursework:** Probability & Statistics for Engineers, Signals & Systems, Search & Data Mining, Computer Networks, Computer Architecture, Digital Signal Processing, Embedded Systems

Experience

Control Systems Engineer Intern | Multi-Dimensional Integration *2025-06 to 2025-08*

- Designed and deployed a FastAPI-based Python tool to automate JSON configuration editing for Ignition SCADA projects, reducing manual workflow time and eliminating a class of human error in config management
- Built a custom Python library for programmatic manipulation of Ignition JSON data structures, enabling repeatable and testable configuration workflows across multiple projects
- Developed interactive Python and Java interfaces within the Ignition platform to improve operator visibility and control over live industrial processes
- Extended legacy control system projects by interpreting engineering drawings and technical documentation, validating outputs against original design specifications
- Collaborated within a multidisciplinary engineering team to scope, implement, and deliver features against project management milestones

Clinical Studies Data System Intern | Johns Hopkins University, Department of Epidemiology *2024-06 to 2024-08*

- Developed JavaScript tooling for clinical trial data entry forms, improving data validation and reducing entry errors across active studies
- Built a PHP program to convert and migrate a Microsoft Access database to a web-compatible structure, preserving referential integrity across thousands of records
- Maintained and improved client-server architecture for web-based data collection forms used in ongoing epidemiological research
- Worked iteratively with researchers and staff to incorporate user feedback into system improvements under real-world data integrity constraints

Lab Assistant – Analog Circuits I & Digital Systems | University of Delaware, Dept. of Electrical & Computer Engineering *2024-02 to 2026-05*

Tech Intern | Mid Atlantic Community Church *2022-05 to 2023-08*

Projects

CradleWave Baby Monitor | *Python, FastAPI, 60GHz Radar, Signal Processing* *github.com/SmileyNiloc/CradleWave*

- Engineered an end-to-end non-contact monitoring system using 60 GHz FMCW radar, extracting real-time vital signs via custom signal processing.
- Developed the supporting full-stack IoT architecture, including a Python/FastAPI backend for real-time data ingestion, Google Firebase integration, and a Vue.js web dashboard for live data review.

Ignition JSON Automation Tool | *Python, FastAPI, REST API, SCADA, Ignition, Automation*

- Built a FastAPI application exposing a REST interface for programmatic JSON editing of Ignition project configurations
- Developed a reusable internal Python library abstracting common Ignition JSON data manipulation patterns
- Reduced configuration turnaround time and introduced testability into previously manual processes

RemAIk – Full-Stack Browser Game | *Vue.js, Firebase, JavaScript, Full-Stack, Authentication*
github.com/SmileyNiloc/remAIk/tree/main

Arduino Shield Cloud Timer | *Arduino, Embedded Systems, PCB Design, IoT, Cloud, C++*

Leadership & Awards

Cru President | Cru at University of Delaware *2023-01 to Present*

- Coordinated event planning with university administration for events of 50+ students
- Grew a collaborative campus community through intentional relationship building and programming

Eagle Scout (*2022-08*) • **Tau Beta Pi – IEEE Honor Society** (*2025-03*)

Skills

Programming Languages: Python, C, C++, JavaScript, Bash, ARMv8 & x86 Assembly, VHDL, HTML, CSS, PHP
Frameworks & Libraries: FastAPI, Vue.js, Node.js, NumPy, Pandas, Typescript, Jython
Systems & Infrastructure: Linux / Ubuntu (daily driver), Raspberry Pi, Google Firebase, Google Cloud, Git, Docker
Hardware & Embedded: VHDL, Vivado, MPLAB, Arduino, PCB Design, Analog Circuits, 60GHz FMCW Radar
Software & Tools: Visual Studio Code, WinSCP, Ignition SCADA, Inkscape, ProPresenter, Proton/Wine